

10 Empirical Mapping Interface

This section leaves audited insertion points for future data. It does not claim that any named measure is currently available, complete, validated, or identifying. Every interface specifies a unit, observational grain, required join keys, availability status, and identification limit.

10.1 Developer and project interfaces

10.1.1 Interface 1: Predetermined research capability a_i

Meaning: pre-existing ability to generate and advance viable original-drug projects. *Candidate measure:* pre-policy development history, innovative-drug pipeline, scientific labor, or patent history after construct validation. *Unit:* capability index or pre-policy event rate. *Grain:* developer by pre-policy period. *Required keys:* stable developer identifier, source identifier, and event or reporting date. *Availability:* NOT VERIFIED; no source is asserted available. *Identification limit:* patent history may proxy predetermined capability, but it is not x_i and does not identify a_i separately without additional structure.

10.1.2 Interface 2: Advancement intensity x_i

Meaning: project-development / advancement intensity.

Candidate measure: validated development-stage activity. *Unit:* latent intensity or investment per developer-period. *Grain:* developer by decision cohort or period. *Required keys:* developer identifier and decision-cohort date, with a documented activity definition. *Availability:* NOT DIRECTLY OBSERVED and NOT VERIFIED. *Identification limit:* x_i advances viable projects toward route planning and excludes upstream patent-generating research. Candidate development-stage measures require construct and timing validation.

10.1.3 Interface 3: Planning-stage projects $a_i x_i$

Meaning: viable projects reaching the route-planning stage. *Candidate measure:* IND or regulatory application activity and early clinical milestones after a measurement audit. *Unit:* projects or qualifying events per developer-period. *Grain:* developer-project-event date. *Required keys:* developer identifier, project or product identifier, event type, and event date. *Availability:* NOT VERIFIED. *Identification limit:* this is not approval or launch and does not separately identify a_i and x_i .

10.1.4 Interface 4: Internal manufacturing capability k_i

Meaning: predetermined ability to manufacture a project's required technology internally. *Candidate measure:* pre-reform, project-compatible capacity evidence. *Unit:* compatibility/capability index or physical capacity in a defined technology class. *Grain:* developer-facility-technology class-date. *Required keys:* developer, facility, technology class, product or requirement class, and observation date. *Availability:* NOT VERIFIED. *Identification limit:* licences or GMP scope alone need not measure usable project-compatible capacity.

10.1.5 Interface 5: Project value q

Meaning: project commercial/scientific value shifter. *Candidate measure:* therapeutic value, target value, indication, or market-size measures after validation. *Unit:* standardized project-value index.

Grain: project–indication–reference date. *Required keys:* project/product, indication, valuation source, and reference date. *Availability:* NOT VERIFIED. *Identification limit:* proxies discipline a value shifter; they do not permit a direct MAH shift in q .

10.1.6 Interface 6: Manufacturing complexity m

Meaning: project-specific manufacturing requirement. *Candidate measure:* modality, dosage form, biologic/sterile status, or audited technical class. *Unit:* technology class or validated complexity index. *Grain:* project/product–manufacturing requirement. *Required keys:* project/product identifier, modality, dosage form, technical class, and classification version. *Availability:* NOT VERIFIED. *Identification limit:* m is a project requirement, not developer capacity and not a policy outcome.

10.2 Route and realized-outcome interfaces

10.2.1 Interface 7: Retained entrusted route $r_i^* = E$

Meaning: the developer remains the holder while a qualified external producer manufactures. *Candidate measure:* product-level holder differs from manufacturer. *Unit:* binary route indicator. *Grain:* product–holder–manufacturer–effective date. *Required keys:* normalized holder identifier, manufacturer identifier, product identifier, authorization/effective date, and source provenance. *Availability:* NOT VERIFIED. *Identification limit:* holder, manufacturer, product and timing must all align; holder–manufacturer difference is not automatically route E .

10.2.2 Interface 8: Realized retained outcome Y_i^{ret}

Meaning: realized innovative product for which the developer retains holder status. *Candidate measure:* approval, launch, or marketing-licence outcome with holder identity. *Unit:* product event or count. *Grain:* developer–product–approval/launch date. *Required keys:* developer/holder, product, event type, event date, and jurisdiction. *Availability:* NOT VERIFIED. *Identification limit:* this is a downstream realized outcome, not a direct observation of advancement.

10.2.3 Interface 9: Novelty class g

Meaning: exogenous original-versus-incremental empirical classifier. *Candidate measure:* verified regulatory classification. *Unit:* categorical label O or Inc. *Grain:* product/project–classification version. *Required keys:* product/project identifier, classification rule, source, version, and classification date. *Availability:* NOT VERIFIED. *Identification limit:* the label is not a control or state, creates no x_{ig} , and imposes no response ranking.

10.2.4 Interface 10: Patent applications

Meaning: candidate patenting-side signal outside the baseline endogenous outcome. It may reflect upstream research activity after construct validation, but it is not a direct observation of upstream research. *Candidate measure:* patent applications or families after assignee and quality audits. *Unit:* application/family count or quality-weighted measure. *Grain:* assignee–patent family–application date. *Required keys:* assignee, developer crosswalk, family identifier, application date, and technology class. *Availability:* NOT VERIFIED. *Identification limit:* a patent response neither equals x_i nor directly identifies upstream research, and it does not alone validate or falsify the commercialization mechanism.

10.3 Market and outside-option interfaces

10.3.1 Interface 11: CMO price or scarcity p_m^*

Meaning: equilibrium scarcity of qualified manufacturing service. *Candidate measure:* service price if observed; otherwise audited capacity, density, utilization, or waiting-time proxies. *Unit:* currency per qualified-capacity unit, or a documented proxy unit. *Grain:* technology/region–market period. *Required keys:* technology class, region, period, capacity unit, and supplier/source identifier. *Availability:* PRICE DATA NOT VERIFIED; alternative proxies also require audit. *Identification limit:* a nonprice scarcity proxy does not separately identify p_m^* or the supply and demand schedules.

10.3.2 Interface 12: Transfer outside option $T(q, m)$

Meaning: non-retained transfer or licence value. *Candidate measure:* future transfer/licence deal moments. *Unit:* currency per project or standardized deal-value index. *Grain:* project–licensor–licensee–deal date. *Required keys:* project, licensor, licensee, deal date, deal scope, and disclosed value components. *Availability:* FUTURE INTERFACE; NOT VERIFIED. *Identification limit:* the outside option is not endogenized in the baseline, and incomplete deal values need not identify $T(q, m)$.

10.4 Timing interface

The theoretical anticipation channel remains

$$M \longrightarrow \text{anticipated availability and value of } E \longrightarrow \Omega_i \longrightarrow x_i^*, \quad (1)$$

whereas the realized sequence remains

$$\begin{aligned} x_i^* &\longrightarrow \text{planning-stage projects} \longrightarrow r_i^* \\ &\longrightarrow \text{observed holder–producer separation} \longrightarrow \text{realized products.} \end{aligned} \quad (2)$$

Future tests may examine both sequences, but an observed downstream separation cannot be placed before the ex ante choice of x_i .

10.5 Interface conclusion

These entries are schemas for future validated data. No empirical data set is asserted to be currently available. Before use, every candidate source must pass provenance, coverage, key-uniqueness, unit, timing, and construct-validity audits. The model may discipline composite moments without separately identifying every primitive.